6. Hydrogels are macromolecules, meaning they contain very large numbers of atoms.

They are polymers made from hydrophilic polymer chains. This means that they readily mix with water.

Cross-links between the polymer chains enable hydrogels to form their three-dimensional shape.

The cross-links that join the individual polymer chains together are formed by small reactive molecules that contain one or more double bonds.

The interesting feature of hydrogels is their ability to swell and shrink, depending on their surroundings. When in contact with water, hydrogels are able to absorb water and swell to many times their original size and weight, whilst maintaining the same shape.

Being able to absorb water like this has enabled hydrogels to be used in a wide range of applications. For many of these applications it is important to understand how the properties of hydrogels change with temperature.

Table 1 and **Table 2** show how temperature affects the properties of hydrogel beads in water over time.

Time (hours)	Mass of bead (g)		
	10 °C	20 °C	40 °C
0	0.035	0.035	0.035
2	1.305	2.980	4.280
4	2.512	3.681	5.904
6	3.613	4.389	6.760
8	4.298	5.195	8.101
10	5.120	5.572	8.101
12	5.400	5.980	8.101

Table 1 – the effect of temperature on the mass of hydrogel beads

Time (hours)	Diameter of bead (mm)		
	10 °C	20 °C	40 °C
0	3.56	3.56	3.56
2	10.30	13.00	18.10
4	13.40	16.50	21.70
6	16.20	18.50	23.60
8	17.80	19.60	25.24
10	19.50	20.80	25.24
12	20.50	22.00	25.24

Table 2 – the effect of temperature on the diameter of hydrogel beads

Examiner only

(a) Tick (✓) the statement that **best** describes the molecules that form the cross-links within hydrogels. [1]

they are macromolecules ☐

they are unsaturated ☐

they contain hydrophilic groups ☐

they are polymers ☐

(b) Give the phrase in the passage that shows the change the hydrogel undergoes is a physical rather than a chemical change. [1]

....

....

(c) Calculate the percentage increase in **mass** of a hydrogel bead when placed in water at 20 °C for 6 hours. [2]

Percentage increase = ... %

(d) Write a conclusion for the effect of **temperature** on the diameter of hydrogel beads in water over time. [2]

....

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